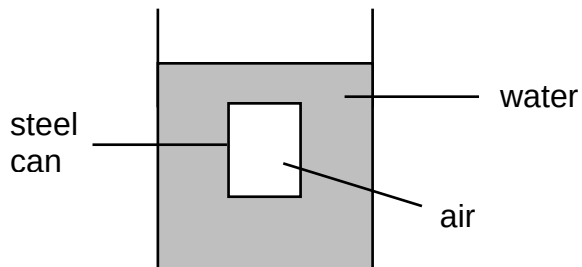


- 5 A sealed steel can filled with air is situated below the surface of water.



What is the origin of the upthrust that acts on the can?

- A** The air pressure in the can is less than the water pressure outside the can.
- B** The water pressure on the bottom of the can is greater than the water pressure on the top.
- C** The air in the can is less dense than the water.
- D** The weight of displaced water acts upwards on the can.

Answer: B

There is higher pressure acting on the lower surface of the steel can, compared to that acting on the upper surface. Due to the pressure difference, there is a net upward force (upthrust) acting on the steel can by the water.

- 6 A uniform plate of weight 6 N is held horizontally by vertical forces. Forces of 4 N act at points